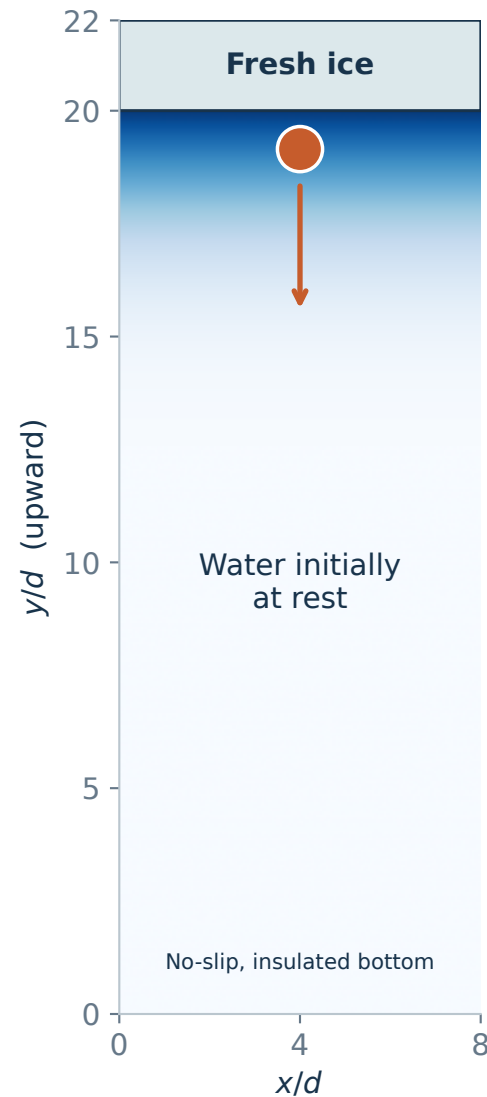


P1b | A grain settling beneath an ice shelf

Scenario schematic · initial condition · one resolved 0.2 mm quartz sphere · melting active



Periodic in x and z

A local snapshot of the meltwater layer

700 dbar · far-field CT = 0 °C · SA = 34.85 g/kg

BOX

$8d \times 22d \times 8d = 1.6 \times 4.4 \times 1.6 \text{ mm}$

INITIAL STATE

Ice interface: $y/d = 20$

Grain centre: $(4, 19.15, 4)d$; top clearance: $0.35d$

Interface water: -1.670 °C , 21.30 g/kg

WHAT WE MEASURE

Settling speed · meltwater carried downward

Wake evolution · scalar transport · continued melting

Schematic is to scale in x-y; shading indicates prescribed meltwater. Release is the initial condition; no background current is imposed.